

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Date	
Team ID	
Project Name	Auto-Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Technical Architecture

The solution is implemented within the ServiceNow platform. The student uses the School IT Tickets interface, ticket records are stored in the custom School IT Ticket table, Flow Designer applies the classification rules, and ServiceNow email, ACLs, roles and Update Sets support notification, security and controlled change tracking.

Technical Architecture - School IT Ticket Automation

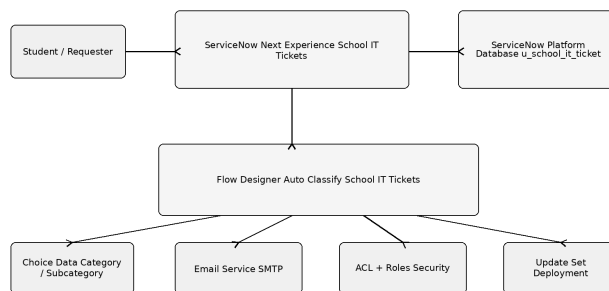


Figure 1: Technical architecture of the ServiceNow-based ticket automation solution.

Technical Architecture Diagram Explanation

System Architecture Breakdown:

- **Presentation Layer:** Students submit requests via the ServiceNow Next Experience School IT Tickets form. Technicians view records through standard list and form views.
- **Database Layer:** Ticket records are stored in the custom table `u_school_it_ticket`, with custom fields for short description, category, subcategory, state, and caller reference.
- **Automation Engine:** Flow Designer executes natively in the platform cloud without external middleware, evaluating condition branches on record insert.
- **Platform Services:** ServiceNow SMTP Email service sends automated notifications, ACLs govern record permissions, and Update Sets handle change management.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interface used to create and view School IT Tickets.	ServiceNow Next Experience
2	Application Logic	Automatic keyword-based classification and notification workflow.	ServiceNow Flow Designer
3	Database	Stores School IT Ticket records and fields.	ServiceNow platform database
4	Cloud Database	No separate cloud database is used by this solution.	Not used
5	File Storage	No application-specific file storage is required.	Not used
6	External API	No third-party API is required for the implemented workflow.	Not used
7	Machine Learning Model	No machine-learning model is used; classifier is rule/keyword based.	Not used
8	Security	Controls access to the custom table and records.	ServiceNow ACLs and Roles
9	Change Tracking	Tracks table, dictionary, choice, flow and related customizations.	ServiceNow Update Sets
10	Infrastructure	Application execution and platform services.	ServiceNow Cloud Platform

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is used in the implemented solution.	ServiceNow platform
2	Security Implementations	The custom ticket table uses access control and role configuration.	ServiceNow ACLs / Roles
3	Scalable Architecture	The event-driven flow can apply the same classification process to multiple ticket records.	Flow Designer
4	Availability	The application is accessed through the ServiceNow platform and School IT Tickets module.	ServiceNow platform
5	Performance	Classification occurs automatically after record creation for supported issue keywords.	Flow Designer conditions/actions

Implementation Evidence

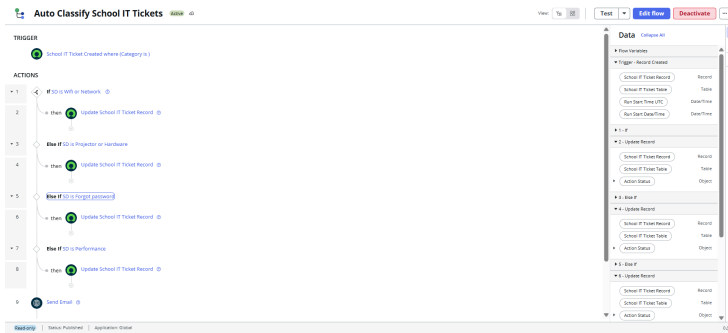


Figure 2: Published Flow Designer workflow.

The image shows a ServiceNow "Update Set - Project Update Set" page. The "Name" field is "Project Update Set", the "Status" is "Complete", and the "Created" date is "2024-09-26 10:17:57". The "Created by" field is "admin". The "Merged to" field is empty. The "Description" field is empty. Below the form, there is a table of "Customer Updates (42)".

Created	Type	View	Target name	Updated by	Remove update set	Action
2024-09-26 10:17:57	Application Menu		School IT Support	admin	Import	IMPORT_OR_UPDATE
2024-09-26 10:17:57	Variable		School IT Tickets	admin	Import	IMPORT_OR_UPDATE
2024-09-26 10:17:57	Choice list		School IT Ticket Category	admin	Import	IMPORT_OR_UPDATE
2024-09-26 10:17:57	Choice list		School IT Ticket State	admin	Import	IMPORT_OR_UPDATE
2024-09-26 10:17:57	Choice list		School IT Ticket Subcategory	admin	Import	IMPORT_OR_UPDATE
2024-09-26 10:17:57	Table		School IT Ticket	admin	Import	IMPORT_OR_UPDATE
2024-09-26 10:17:57	Dictionary		School IT Ticket	admin	Import	IMPORT_OR_UPDATE
2024-09-26 10:17:57	Dictionary		School IT Ticket Assigned Group	admin	Import	IMPORT_OR_UPDATE

Figure 3: Completed Project Update Set with captured customer updates and XML export link.

Deployment & Packaging Architecture

All schema definitions, choice list dependencies, form layouts, Flow Designer actions, and notification triggers created during development were captured inside a dedicated ServiceNow Update Set. Upon verification, the Update Set was marked as Complete and exported as an XML package for version-controlled backup and deployment across environments.